PHYS5033

COUNCIL SUSTAINABILITY REPORT

A personal consumption footprint and practical action for the City of Sydney

Prepared for: [Sustainability Manager / Council name]

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Contents

1. Introduction

2. Methodology

3. Results

4. Discussion

5. Recommendations

6. Conclusion

References

Supplementary Information

1. Introduction

Reducing greenhouse gas emissions is necessary to limit further warming and protect livelihoods, ecosystems and infrastructure. Human activities have unequivocally caused global warming, and decisions made during this decade affect the risks experienced by future generations. The Intergovernmental Panel on Climate Change (IPCC, 2023) identifies rapid and sustained emissions reductions as central to limiting those risks. This requires changes across energy, transport, food and material consumption, rather than attention to a single visible source.

Australia's Climate Change Act 2022 establishes targets of reducing net emissions by 43% below 2005 levels by 2030 and achieving net zero by 2050 (Australian Government, 2022). National progress depends on energy and industrial changes, but also on the services, infrastructure and consumption choices that shape everyday demand. For a resident of Redfern, these choices include food purchases, apartment electricity and transport. The City of Sydney can help residents act through accessible information, local facilities and cooperation with housing providers and transport authorities.

A personal footprint assessment makes these connections measurable. An electricity bill reveals one part of the picture, while food, clothing and services also require production and energy elsewhere. Input output (IO) analysis follows the economic links between suppliers and customers, allowing purchases to be connected to emissions throughout their supply chains (Miller & Blair, 2009). This helps distinguish the largest expenses from the activities with the greatest emissions consequences.

This report evaluates the annual consumption footprint of an illustrative Redfern resident and identifies practical actions for the resident and the council. It presents the total footprint, examines consumption categories and emitting sectors, and places the result in an Australian and international context. A food substitution scenario then tests an affordable response to the principal findings.

2. Methodology

2.1 Resident profile and expenditure

The assessment represents one adult sharing an apartment with another adult, without a private car and mainly using public transport. Realistic hypothetical expenditure covers November 2025 to October 2026. Shared bills are divided equally, with the resident's share recorded directly. No flights, paid accommodation, major appliance purchases or tuition expenses are assumed during this period.

The annual budget is AUD 29,760, including rent of AUD 15,600, equivalent to AUD 300 per week. Monthly amounts are multiplied by 12. Twenty expenditure items are divided into 40 allocations and mapped to the original 345 IO sectors. For example, groceries are divided among food products, while public transport is divided between passenger rail and bus or tram services. The allocations sum to the original budget; unused sectors remain in the vector as zeros. Supporting records are included in Supplementary Information S1–S3.

2.2 IO model and footprint equations

The supplied Australian teaching dataset describes the economy in 2018 (The University of Sydney, n.d.). T is the 345 × 345 intermediate transaction matrix, x the sector output vector and Q the emissions account. T and x are measured in AUD million, while Q is measured in kilotonnes of carbon dioxide equivalents, or CO_2e. The model follows the standard Leontief framework (Miller & Blair, 2009):

$$A = T\,\hat{X}^{-1}, \quad A_{ij} = T_{ij} / x_j; \quad L = (I - A)^{-1} \quad (1)$$

Here $\hat{X}$ is the diagonal matrix of output and I is the identity matrix. A describes the direct inputs needed for each unit of sector output. L, the Leontief inverse, captures those inputs plus the successive production requirements of upstream suppliers.

$q = Q\,\hat{x}^{-1}$; $m = qL$; $F = qLy = my$ (2)

The row vector q measures direct emissions per unit of output. Multiplication by L gives total intensity m, including the supply chain. The personal demand vector y contains annual expenditure in AUD million. F is obtained in kt CO_2e and multiplied by 1,000 to report tonnes per person per year. Excel calculations use MUNIT, MINVERSE and MMULT, as documented in Supplementary Information S4–S5.

2.3 Decomposition, checks and food scenario

Consumption contributions are calculated as $c_j = m_j y_j$. Emitting sector contributions are $e_i = q_i(Ly)_i$. Each decomposition sums to F: the first identifies purchases driving emissions and the second identifies where emissions occur. Checks confirm sector order, units, expenditure totals and the inverse relationship between I − A and L.

The food scenario reduces expenditure on fresh meat, poultry, milk and cheese by 20%. The released AUD 384 is allocated equally to legumes and vegetables. Total expenditure and the grocery budget remain unchanged. Replacing y with the revised vector y' gives $F' = qLy'$, with all model intensities held constant. The scenario is recorded in the Reduction scenario worksheet and Supplementary Information S7.

2.4 National references and interpretation

National references use consumption CO_2 per person from Our World in Data, based on Global Carbon Budget 2025 emissions and population estimates (Global Carbon Project, 2025; Our World in Data, n.d.). The primary year is 2018, matching the IO reference year. Historical lines cover 1990–2023. The selected countries provide contrasting contexts rather than an exhaustive ranking.

2.5 Assumptions and limitations

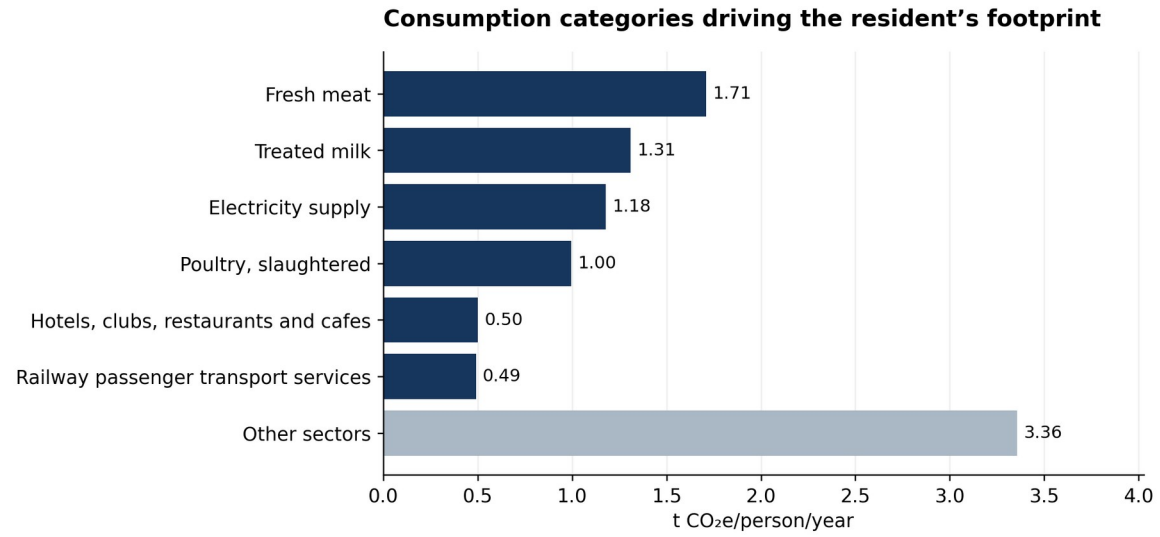
The IO model uses average sector intensities and fixed production relationships, so it does not distinguish individual brands or suppliers. Nominal 2025–26 spending is applied to 2018 intensities without deflation; taxes, trade margins and insurance service charges are not separately adjusted. Household gas combustion is not added as a separate estimate. These choices define the reported consumption footprint.

National data cover fossil and industrial CO_2 adjusted for trade, excluding other greenhouse gases and land use change. They also represent wider national final demand rather than only private purchases. Therefore, the national references provide context for the resident's CO_2e estimate, not a directly comparable greenhouse gas ranking. The scenario measures a change within the same model, not an observed outcome. These qualifications are applied throughout the interpretation without changing the underlying calculations.

3. Results

3.1 Total footprint and consumption contributions

The resident's annual footprint is 9.54 t CO_2e. Groceries and meals purchased outside the home together contribute 6.06 t, or 63.5% of the total. Electricity contributes 1.18 t, or 12.4%, while public transport and occasional taxi travel contribute 0.87 t, or 9.1%.



Source: Labels.xlsx. "Other" sums the remaining consumption categories.

Figure 1. Annual footprint by consumption category. "Other" combines the remaining categories. Source: calculations from the supplied IO dataset (The University of Sydney, n.d.).

Fresh meat is the largest individual purchase category at 1.71 t, followed by milk at 1.31 t, electricity at 1.18 t and poultry at 1.00 t. Together, these four categories account for 54.4% of the footprint. Meals outside the home contribute 0.50 t and passenger rail contributes 0.49 t. The "Other" bar aggregates many smaller categories rather than identifying one additional priority.

The pattern differs sharply from expenditure. Rent accounts for 52.4% of the budget but contributes 0.28 t, or 2.9% of the footprint. Groceries and dining account for 24.2% of spending yet nearly two thirds of emissions. Ranking expenses alone would therefore miss the main opportunities.

3.2 Emitting sectors

The emitting sector view identifies beef cattle production at 2.23 t, electricity supply at 1.36 t and treated milk production at 1.21 t. These results locate important sources upstream of the resident's purchases. For example, electricity supports food and service production as well as the apartment. This is a second perspective on the same 9.54 t total; its contributions are not added to the consumption categories.

3.3 Australian and international context

Australia's 2018 consumption CO_2 reference is 15.38 t per person. The United States records 16.97 t, Japan 10.04 t, the United Kingdom 8.17 t, China 6.66 t and India 1.63 t (Global Carbon Project, 2025; Our World in Data, n.d.). Figure 2 places the resident's 9.54 t CO_2e result alongside these national references using the definitions established in Section 2.5. Australia lies near the upper end of the selected national series.

Annual emissions per person: resident and national references

National values for 2018, with historical consumption trends.

PERSONAL CONSUMPTION ESTIMATE

Illustrative Redfern resident	**9.54 t CO_2e/person/year**

Annual spending: 2025–26. Economic model: 2018.

NATIONAL CONSUMPTION REFERENCES (CO_2)

Country	2018: t CO_2/person	Trend: 1990–2023
United States	**16.97**	
Australia	**15.38**	
Japan	**10.04**	
United Kingdom	**8.17**	
China	**6.66**	
India	**1.63**	

Country data cover fossil and industrial CO_2 adjusted for trade.

Source: Our World in Data / Global Carbon Budget (2025); accessed 8 November 2026.

Personal results are CO_2e; national references are CO_2. Definitions are given in Methodology.

Sparklines use individual vertical scales; dots mark 2018.

Figure 2. Resident footprint and national consumption references. Country values are for 2018; dots mark 2018 on the 1990–2023 trends, which use individual vertical scales. Sources: Global Carbon Project (2025) and Our World in Data (n.d.).

3.4 Food substitution scenario

The revised food basket reduces the footprint from 9.54 to 9.02 t CO_2e per year. The saving is 0.52 t, equivalent to 5.5% of the baseline total, while annual expenditure remains AUD 29,760. This demonstrates a reduction through changing the composition of purchases rather than increasing the budget.

4. Discussion

4.1 What the assessment reveals

The assessment shifts attention from visible activities to the demand supporting them. A resident without a car might initially focus on electricity and public transport. Those remain important, but the results show that food is the stronger priority. Fresh meat and dairy purchases connect the shopping basket to agricultural emissions, while electricity appears across many supply chains. This explains why a footprint extends beyond emissions released inside the home.

The distinction between expenditure and impact also matters for practical advice. Rent dominates the budget, but food dominates the modelled footprint. Asking the resident simply to spend less would provide little guidance about which changes matter. The scenario instead identifies a specific reallocation within the existing grocery budget. It provides a measurable starting point without assuming that the resident can afford expensive replacement equipment.

The two decompositions suggest complementary responsibilities. Consumers influence the demand for products, while farms, manufacturers and electricity suppliers influence emissions during production.

Personal choices therefore work alongside changes in production systems. Councils can assist residents, but cannot replace the wider transition in electricity and industrial supply chains.

4.2 Why this matters for sustainability

Australia's position among the national references gives local consumption a wider significance. Resident demand draws on production beyond council boundaries, so a local assessment should consider imported goods and upstream activity as well as visible emissions. At the same time, residents need accessible alternatives rather than numerical comparisons alone. The purpose of the assessment is to identify action, not to assign personal blame.

Sustainability also requires attention to affordability, health and participation. The IPCC (2023) emphasises the relationship between climate action and sustainable development. A response that reduces emissions but makes adequate food or essential services harder to obtain would be incomplete. The proposed food changes should respect nutritional needs and preferences, while energy measures should maintain comfortable living conditions. Essential healthcare and housing should not be cut simply to lower a spending footprint.

The shared apartment is especially relevant to implementation. The resident can change meal planning and electricity use, but appliance replacement, building improvements and solar installation may require a landlord or owners corporation. Public transport is already the principal travel mode, making preservation of that pattern more useful than advice to give up a car the resident does not own. Council support should address these practical conditions.

5. Recommendations

5.1 Actions for the resident

Food should be the first priority. Over the next three months, the resident should trial the scenario's 20% reduction in expenditure on fresh meat, poultry, milk and cheese, directing the AUD 384 annual equivalent towards legumes and vegetables. A weekly meal plan can make these substitutions deliberate and affordable. Receipts should be recorded by the same categories used in the assessment, while food preferences and nutritional adequacy guide the final choices. The modelled 0.52 t saving provides a benchmark for the trial.

Meal planning should also reduce avoidable waste. Checking existing supplies before shopping, storing food appropriately and using leftovers can prevent purchases that deliver no nutritional benefit. Dining out need not be eliminated, but the resident should consider meal choice and portion size when reviewing the AUD 2,400 annual dining budget.

Electricity is the next priority. The resident should compare household electricity use with a consistent baseline and aim for a 10% reduction in kWh over the following year. Turning off unused equipment, operating full laundry loads and managing heating or cooling sensibly are accessible starting points. Both occupants should agree on changes and keep the same allocation method when reviewing their bills. Larger upgrades should be discussed with the landlord rather than assumed to be within the resident's control.

The resident should retain public transport as the main travel mode and use walking for suitable short trips. Occasional taxi journeys should be reviewed for trips that can reasonably be combined or replaced. Clothing and household goods are secondary priorities: repairing, borrowing and buying used items can reduce demand for new production without diverting attention from food and electricity.

5.2 Actions for the City of Sydney

The council should organise practical sessions on affordable meal planning and cooking with legumes and vegetables through local community facilities. These should translate footprint findings into realistic shopping choices, with examples suitable for small shared kitchens. Working with community organisations and food retailers would help reach residents with different incomes and dietary needs. Evaluation should measure participation and reported changes in purchases rather than attendance alone.

For energy, the council should provide advice designed for tenants and apartment residents, including simple checks of electricity bills and guidance on discussing improvements with landlords and owners corporations. Advice should separate actions residents can take immediately from changes requiring building approval. This would help turn the 1.18 t electricity finding into achievable household decisions.

The council should support the existing transport pattern through safe pedestrian routes, useful bicycle connections and clear access information around local destinations. Where improvements involve rail or bus services, it should coordinate with the responsible transport authority rather than assume control over routes or fares. Repair and reuse activities can complement these priorities by making access to durable goods more affordable.

A twelve month pilot should begin with food and electricity because they offer the clearest links to this assessment. At the end of the year, participating residents could repeat the expenditure exercise, record electricity use and report barriers encountered. The council should use these findings to improve support and communicate aggregate progress without collecting unnecessary personal financial details.

6. Conclusion

The resident's annual footprint is 9.54 t CO_2e, with food contributing almost two thirds of the total and electricity providing another clear priority. Sector analysis connects these purchases to agricultural and energy production, showing why expenditure totals alone are insufficient for choosing action. National consumption references place the assessment within Australia's wider emissions context. A food reallocation within the existing budget lowers the modelled footprint to 9.02 t, a 5.5% reduction. The most useful response is therefore a combination of specific resident choices and council support that makes those choices practical. Affordable food initiatives, advice for apartment residents and accessible transport can turn the assessment into sustained local action.

References

Australian Government. (2022). *Climate Change Act 2022 (Cth)*. Federal Register of Legislation. https://www.legislation.gov.au/C2022A00037/latest/text

Global Carbon Project. (2025). *Global Carbon Budget 2025* [Data set]. https://globalcarbonbudget.org/

Intergovernmental Panel on Climate Change. (2023). *Climate change 2023: Synthesis report*. https://doi.org/10.59327/IPCC/AR6-9789291691647

Miller, R. E., & Blair, P. D. (2009). *Input–output analysis: Foundations and extensions* (2nd ed.). Cambridge University Press. https://doi.org/10.1017/CBO9780511626982

Our World in Data. (n.d.). *CO_2 and greenhouse gas emissions* [Data set and codebook]. Retrieved November 8, 2026, from https://github.com/owid/co2-data/tree/382ee6c662b0ece26e111f263b44c029afad7787

The University of Sydney. (n.d.). *PHYS5033: Australian 345 sector input output teaching dataset (2018 economy)* [Course supplied data]. Labels.xlsx.

Supplementary Information

S1. Annual personal expenditure

All 20 spending items, monthly-to-annual calculation and total; source: Annual expenditure. Worksheet snapshots rendered by LibreOffice Calc from Labels.xlsx.

S1. Personal annual expenditure

Source: Annual expenditure!A1:G22; full baseline, personal shares already allocated.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Row	Expenditure item	Monthly AUD	Months	Annual AUD	Basis
2	Shared apartment rent	1,300.000	12.000	15,600.000	Monthly × 12
3	Groceries and non-alcoholic drinks	400.000	12.000	4,800.000	Monthly × 12
4	Restaurants, cafes and takeaway	200.000	12.000	2,400.000	Monthly × 12
5	Electricity	40.000	12.000	480.000	Monthly × 12
6	Household gas	10.000	12.000	120.000	Monthly × 12
7	Water	10.000	12.000	120.000	Monthly × 12
8	Home internet	30.000	12.000	360.000	Monthly × 12
9	Mobile phone service	30.000	12.000	360.000	Monthly × 12
10	Public transport	150.000	12.000	1,800.000	Monthly × 12
11	Taxi and rideshare	20.000	12.000	240.000	Monthly × 12
12	Clothing and footwear	50.000	12.000	600.000	Monthly × 12
13	Small household goods and cleaning products	30.000	12.000	360.000	Monthly × 12
14	Toiletries and personal-care products	40.000	12.000	480.000	Monthly × 12
15	Medical, dental and pharmaceutical expenses	50.000	12.000	600.000	Monthly × 12
16	Recreation and entertainment services	50.000	12.000	600.000	Monthly × 12
17	Books and learning materials	30.000	12.000	360.000	Monthly × 12
18	Contents insurance	20.000	12.000	240.000	Monthly × 12
19	Hairdressing	20.000	12.000	240.000	Monthly × 12
20	Flights and accommodation	0.000	12.000	0.000	Monthly × 12
21	Vehicle purchase, fuel and maintenance	0.000	12.000	0.000	Monthly × 12
	TOTAL			29,760.000	AUD/person/year

Hypothetical Redfern adult; two-adult shared apartment; no private car. Rent AUD 300/week = 15,600/year. Shared costs are personal shares, not household totals.

S2a. Expenditure mapping — part 1

First 20 allocation rows; source: Expenditure split. All shares are illustrative assumptions. Worksheet snapshots rendered by LibreOffice Calc from Labels.xlsx; not Microsoft Excel window captures.

S2. Mixed-expenditure allocation

Source: Expenditure split!A1:I41. All 40 allocation rows shown; percentages are explicit scenario assumptions.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Source row	Split component	Sector ID	Share	Annual AUD	IO sector
2.000	Rent maps to dwelling services, not building construction or real-estate agent fees.		292.000	100%	15,600.000 Ownership of dwellings
3.000	Fresh meat		75.000	15%	720.000 Fresh meat
3.000	Poultry		77.000	10%	480.000 Poultry, slaughtered
3.000	Milk		79.000	10%	480.000 Treated milk
3.000	Cheese		80.000	5%	240.000 Cheese
3.000	Vegetables		15.000	15%	720.000 Vegetables
3.000	Fruit		16.000	10%	480.000 Fruit
3.000	Bread		96.000	10%	480.000 Bread and bread rolls
3.000	Pasta		94.000	5%	240.000 Pasta
3.000	Rice products		86.000	5%	240.000 Rice products
3.000	Other food		98.000	10%	480.000 Food products
3.000	Soft drinks		105.000	5%	240.000 Soft drinks
4.000	Purchased meals; no separate ingredient mapping.		259.000	100%	2,400.000 Hotels, clubs, restaurants and cafes
5.000	Electricity supply		241.000	100%	480.000 Electricity supply
6.000	Gas supply		242.000	100%	120.000 Gas supply
7.000	Water/sewerage services		243.000	100%	120.000 Water supply; sewerage and drainage services
8.000	Domestic home internet service		282.000	100%	360.000 Domestic telecommunication services
9.000	Mobile service only		282.000	100%	360.000 Domestic telecommunication services
10.000	Passenger rail		265.000	60%	1,080.000 Railway passenger transport services
10.000	Bus/tram/light rail		262.000	40%	720.000 Bus and tramway

S2b. Expenditure mapping — part 2

Remaining 20 allocation rows; source: Expenditure split. Both parts together show all 40 allocations. Worksheet snapshots rendered by LibreOffice Calc from Labels.xlsx.

S2. Mixed-expenditure allocation

Source: Expenditure split!A1:I41. All 40 allocation rows shown; percentages are explicit scenario assumptions.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Source row	Split component	Sector ID	Share	Annual AUD	IO sector
11.000	Taxi/rideshare proxy		263.000	100%	240.000 Taxi and hire car
12.000	Clothing		118.000	80%	480.000 Clothing
12.000	Footwear		119.000	20%	120.000 Footwear
13.000	Cleaning products		149.000	50%	180.000 Soap and other detergents
13.000	Household paper products		130.000	25%	90.000 Paper products
13.000	Small plastic household goods		160.000	25%	90.000 Plastic products
14.000	Toiletry products		150.000	75%	360.000 Cosmetics and toiletry preparations
14.000	Soap/detergents for personal use		149.000	25%	120.000 Soap and other detergents
15.000	GP/dental/optometry services		322.000	50%	300.000 GPs, dentists, optometrists, ambulance
15.000	Pharmaceuticals		148.000	50%	300.000 Pharmaceutical goods for human use
16.000	Cinema		326.000	40%	240.000 Motion pictures
16.000	Pay-TV/streaming proxy; imperfect sector match		279.000	30%	180.000 Overseas communication services and Pay TV
16.000	Sport/recreation services		337.000	30%	180.000 Sport and recreation services (incl horse and dog racing, sports grounds, services)
17.000	Books		137.000	80%	288.000 Books, maps, magazines
17.000	Stationery		131.000	20%	72.000 Printing and stationery
18.000	Contents insurance: premium used as provisional service-spending proxy; insurance valuation unresolved.		285.000	100%	240.000 Insurance
19.000	Hairdressing personal services		338.000	100%	240.000 Personal services
20.000	Zero baseline; placeholder allocation only. Add accommodation split if future travel scenario changes.		269.000	50%	0.000 Air and space transport
20.000	Zero baseline accommodation placeholder		260.000	50%	0.000 Accommodation
21.000	Zero baseline; fuel placeholder only. Rebuild vehicle purchase/maintenance splits if profile changes.		139.000	100%	0.000 Petrol and diesel

Check source allocation shares total 100%. The full purchaser-price mapping remains provisional: taxes, margins and 2018 price conversion are not resolved.

S3. Final-demand vector

Nonzero sectors only are displayed; IO_345 demand retains all 345 sectors, including zero entries. Worksheet snapshots rendered by LibreOffice Calc from Labels.xlsx.

S3. 345-sector final demand

Source: IO_345 demand!A1:G347. Nonzero baseline sectors shown; zero-demand sectors remain in the full 345-row vector.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Sector ID	Sector name	Annual AUD	y (AUD million)	x (AUD million)	Q (kt CO2-e)	
15	Vegetables	720.000	0.000720	5,215.800	1,177.800	
16	Fruit	480.000	0.000480	4,085.100	936.390	
75	Fresh meat	720.000	0.000720	14,177.000	1,289.100	
77	Poultry, slaughtered	480.000	0.000480	4,463.700	972.620	
79	Treated milk	480.000	0.000480	105.610	266.390	
80	Cheese	240.000	0.000240	2,703.500	301.560	
86	Rice products	240.000	0.000240	463.180	41.096	
94	Pasta	240.000	0.000240	704.600	67.935	
96	Bread and bread rolls	480.000	0.000480	3,877.700	310.560	
98	Food products	480.000	0.000480	7,564.100	796.660	
105	Soft drinks	240.000	0.000240	5,854.000	629.070	
118	Clothing	480.000	0.000480	3,227.200	1,363.900	
119	Footwear	120.000	0.000120	933.030	437.140	
130	Paper products	90.000	0.000090	2,830.000	346.210	
131	Printing and stationery	72.000	0.000072	3,628.400	308.880	
137	Books, maps, magazines	288.000	0.000288	2,536.700	46.136	
148	Pharmaceutical goods for human use	300.000	0.000300	9,899.600	2,035.100	
149	Soap and other detergents	300.000	0.000300	359.570	71.377	
150	Cosmetics and toiletry preparations	360.000	0.000360	857.680	120.920	
160	Plastic products	90.000	0.000090	10,427.000	1,135.900	
241	Electricity supply	480.000	0.000480	43,542.000	60,073.000	
242	Gas supply	120.000	0.000120	3,772.900	0.020	
243	Water supply; sewerage and drainage services	120.000	0.000120	29,893.000	14,851.000	
259	Hotels, clubs, restaurants and cafes	2,400.000	0.002400	61,016.000	195.050	
262	Bus and tramway	720.000	0.000720	7,742.000	1,647.400	
263	Taxi and hire car	240.000	0.000240	5,771.900	2,386.500	
265	Railway passenger transport services	1,080.000	0.001080	7,549.000	2,395.800	
279	Overseas communication services and Pay TV	180.000	0.000180	5.862	0.000	
282	Domestic telecommunication services	720.000	0.000720	44,642.000	152.140	
285	Insurance	240.000	0.000240	40,356.000	106.800	
292	Ownership of dwellings	15,600.000	0.015600	250,390.000	35.241	
322	GPs, dentists, optometrists, ambulance	300.000	0.000300	38,991.000	147.190	
326	Motion pictures	240.000	0.000240	6,381.500	21.279	
337	Sport and recreation services (incl horse and dog racing, sports grounds, services)	180.000	0.000180	19,177.000	73.222	
338	Personal services	240.000	0.000240	46,294.000	173.430	
	TOTAL	29,760.000	0.029760			

y = annual AUD / 1,000,000. T and x are AUD million; Q is kt CO2-e. Matrix/vector order follows row labels exactly.

S4. Matrix construction

Actual MUNIT/MINVERSE formulas and labelled excerpts of $\hat{X}$, $\hat{X}^{-1}$, A and L. Worksheet snapshots from Microsoft Excel.

A1 {=MUNIT(345)*Output!A1:A345}

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	3319.8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	2264.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	1704.8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	7567.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	2255.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	235.41	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	1439.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	838.02	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	9001.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	4161.3	0	0	0	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0	0	637.91	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	965.79	0	0	0	0	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0	0	0	0	1263.5	0	0	0	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0	0	0	0	0	640.73	0	0	0	0	0	0	0	0	0
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5215.8	0	0	0	0	0	0	0	0
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4085.1	0	0	0	0	0	0	0
17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1649.8	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	286.6	0	0	0	0	0
19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	657.99	0	0	0	0
20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	681.9	0	0	0
21	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	90.592	0	0
22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	460.42	0
23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1280.6
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
25	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
27	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

... | IO_345 demand | IO_345 table | Validation | X ^ | X ^ inverse | A coefficients | L matrix | q matrix | m matrix ...

A1 {=MINVERSE(X ^!A1:MG345)}

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	0.000301	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0.000442	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0.000587	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0.000132	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0.000443	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0.004248	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0.000695	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0.001193	0	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0.000111	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0.00024	0	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0	0	0.001568	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0.001035	0	0	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0	0	0	0	0.000791	0	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0	0	0	0	0	0.001561	0	0	0	0	0	0	0
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.000192	0	0	0	0	0	0
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.000245	0	0	0	0	0
17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.000606	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.003489	0	0	0
19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00152	0	0
20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.001466	0
21	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.011039
22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
25	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
27	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

... | IO_345 demand | IO_345 table | Validation | X ^ | X ^ inverse | A coefficients | L matrix | q matrix | m matrix ...

B2 =IntermediateDemand!A1/Output!A1

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	A_ij = T_ij / x_j; equivalent to A = T * Xhat inverse; dimen	1: Sheep a	2: Shorn w	3: Oats, sc	4: Wheat	5: Barley	6: Rice	7: Oilseed	8: Legume	9: Beef cat	10: Untree	11: Dairy c	12: Pigs	13: Poultr	14: Eggs	15: Vegeta	16: Fruit	17: Plant r	18: Flow
2	1: Sheep and lambs	0.002492	0.001924	0.00211	0.002091	0.002125	0.000411	0.002359	0.00208	0.002562	0.002495	0.002748	0.000531	0.000521	0.000453	0.001255	0.001266	0.001516	0.00169
3	2: Shorn wool	0.000451	0.000349	0.000458	0.000379	0.000385	0.000135	0.000427	0.000377	0.000464	0.000453	0.000501	0.000235	0.00023	0.0002	0.000561	0.000566	0.000678	0.00075
4	3: Oats, sorghum and other cereal grains	0.021384	0.016336	0.017516	0.017725	0.018136	0.024776	0.020135	0.017706	0.021957	0.021203	0.023348	0.014229	0.013945	0.012102	0.008614	0.008544	0.010162	0.01125
5	4: Wheat	0.051639	0.03951	0.039648	0.042814	0.04379	0.07588	0.048656	0.042835	0.053066	0.051284	0.056527	0.002277	0.002226	0.001923	0.008332	0.008482	0.009972	0.01117
6	5: Barley	0.006924	0.005416	0.005737	0.005756	0.005867	0.074228	0.006513	0.005743	0.007107	0.00688	0.007571	0.02129	0.020871	0.018212	0.003365	0.003101	0.003693	0.00407
7	6: Rice	7.49E-05	6.99E-05	5.87E-05	6.24E-05	6.36E-05	0.007265	7.07E-05	6.21E-05	7.7E-05	7.46E-05	8.21E-05	9.51E-06	9.32E-06	8.09E-06	2.23E-05	2.25E-05	2.7E-05	3.02E-0
8	7: Oilseeds	0.009008	0.006851	0.006955	0.007455	0.007619	0.001034	0.00846	0.007448	0.009242	0.00892	0.009824	0.000176	0.000172	0.00015	0.001757	0.001783	0.002125	0.00235
9	8: Legumes	0.015049	0.011525	0.011206	0.012472	0.012761	0.033661	0.014175	0.012463	0.015457	0.01492	0.01643	0.006855	0.006703	0.005804	0.001069	0.000985	0.001172	0.00130
10	9: Beef cattle	0.004753	0.003666	0.00425	0.003984	0.004051	0.001019	0.004496	0.003966	0.004886	0.004757	0.005249	0.001432	0.001403	0.001219	0.003393	0.003423	0.0041	0.00458
11	10: Untreated milk	0.001187	0.00092	0.001211	0.000999	0.001014	0.000418	0.001125	0.000994	0.001222	0.001193	0.001321	0.000643	0.00063	0.000547	0.001497	0.00151	0.001809	0.00202
12	11: Dairy cattle	0.000325	0.000251	0.000311	0.000272	0.000277	0.000191	0.000307	0.000271	0.000334	0.000325	0.00036	0.000182	0.000178	0.000155	0.000324	0.000326	0.000391	0.00043
13	12: Pigs	0.000207	0.00016	0.000209	0.000174	0.000177	0.000117	0.000196	0.000173	0.000213	0.000208	0.00023	0.000137	0.000134	0.000116	0.000253	0.000255	0.000305	0.00034
14	13: Poultry	0.000245	0.00019	0.00025	0.000206	0.000209	0.000138	0.000232	0.000205	0.000252	0.000246	0.000272	0.00678	0.006653	0.005771	0.000408	0.000313	0.000375	0.0004
15	14: Eggs	0.000103	7.95E-05	0.00011	8.63E-05	8.76E-05	2.11E-05	9.72E-05	8.59E-05	0.000106	0.000103	0.000114	6.1E-05	5.97E-05	5.19E-05	0.000153	0.000155	0.000185	0.00020
16	15: Vegetables	0.001173	0.00091	0.002134	0.000988	0.001003	0.000263	0.001113	0.000983	0.001208	0.00118	0.001307	0.000696	0.000682	0.000593	0.005741	0.005795	0.006946	0.00773
17	16: Fruit	0.000891	0.000691	0.000952	0.00075	0.000761	0.000185	0.000845	0.000746	0.000917	0.000896	0.000993	0.000528	0.000518	0.00045	0.001314	0.001326	0.001589	0.00177
18	17: Plant nurseries	0.000519	0.000402	0.00137	0.000437	0.000443	0.000107	0.000492	0.000435	0.000534	0.000522	0.000578	0.000309	0.000303	0.000263	0.004315	0.004375	0.005237	0.00582
19	18: Flowers	0.000142	0.00011	0.000381	0.00012	0.000122	2.88E-05	0.000135	0.000119	0.000147	0.000143	0.000159	8.5E-05	8.33E-05	7.23E-05	0.001216	0.001233	0.001475	0.00163
20	19: Grapes for wine	0.00069	0.000535	0.00074	0.000581	0.00059	0.000141	0.000655	0.000578	0.000711	0.000694	0.000769	0.000411	0.000403	0.00035	0.001028	0.001037	0.001243	0.0013
21	20: Horses	0.000199	0.000155	0.000533	0.000168	0.00017	3.77E-05	0.000189	0.000167	0.000205	0.0002	0.000222	0.021303	0.020935	0.018287	0.002026	0.001736	0.002077	0.002
22	21: Deer	7.18E-06	5.58E-06	2.42E-05	6.05E-06	6.14E-06	2.49E-06	6.81E-06	6.02E-06	7.39E-06	7.22E-06	8E-06	4.22E-05	4.15E-05	4.33E-05	8.51E-05	8.66E-05	0.000103	0.00012
23	22: Animal breeding	0.000826	0.000645	0.000792	0.000699	0.000706	9.98E-05	0.000785	0.000693	0.000851	0.000832	0.000918	0.013457	0.013212	0.011511	0.001015	0.000829	0.000992	0.00110
24	23: Sugar cane	0.000368	0.000285	0.000987	0.000309	0.000314	7.7E-05	0.000349	0.000308	0.000378	0.00037	0.00041	0.000584	0.000573	0.000499	0.003145	0.003185	0.003807	0.00422
25	24: Unginned cotton	0.000183	0.000142	0.000197	0.000154	0.000156	3.91E-05	0.000173	0.000153	0.000188	0.000184	0.000203	0.000187	0.000183	0.00016	0.000277	0.000278	0.000334	0.00037
26	25: Hops	6.11E-05	4.78E-05	5.48E-05	5.17E-05	5.22E-05	8.26E-05	5.8E-05	5.13E-05	6.29E-05	6.16E-05	6.8E-05	0.00014	0.000138	0.00012	4.58E-05	4.44E-05	5.32E-05	5.97E-0
27	26: Grass seed	0.001567	0.001203	0.002396	0.001309	0.001332	0.000167	0.001478	0.00131	0.001612	0.001568	0.001745	0.001002	0.000981	0.000851	0.005524	0.005593	0.006703	0.00747
28	27: Hay	0.041451	0.032402	0.036062	0.03514	0.035525	0.004677	0.039412	0.034827	0.042688	0.041884	0.0463	0.024272	0.023789	0.020661	0.02479	0.0248	0.029725	0.03324
29	28: Natural rubber	0.001554	0.001195	0.001501	0.001299	0.001321	0.000167	0.001466	0.001294	0.001598	0.001555	0.001723	0.004536	0.004454	0.003879	0.001687	0.001645	0.001972	0.00219
30	29: Skins and other agricultural services	0.06213	0.048137	0.068378	0.052289	0.053099	0.006588	0.058908	0.052039	0.063963	0.062497	0.069245	0.039581	0.038792	0.033694	0.100169	0.101082	0.121148	0.13548

... | IO_345 demand | IO_345 table | Validation | X ^ | X ^ inverse | A coefficients | L matrix | q matrix | m matrix ...

LS319 | =MINVERSE(MUNIT(345)-'A coefficients'!B2:MH346)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	L = MINVERSE(MUNIT(345) - A); dimensionless; Excel array formula	1: Sheep a	2: Shorn w	3: Oats, sc	4: Wheat	5: Barley	6: Rice	7: Oilseed	8: Legume	9: Beef cat	10: Untrea	11: Dairy c	12: Pigs	13: Poultr	14: Eggs	15: Vegeta	16: Fruit	17: Plant
2	1: Sheep and lambs	1.003606	0.002783	0.003105	0.003021	0.003072	0.001784	0.003411	0.003015	0.003708	0.003609	0.003987	0.002004	0.001962	0.001695	0.002074	0.002079	0.0024
3	2: Shorn wool	0.001158	1.000898	0.00104	0.000974	0.000988	0.000843	0.001097	0.000974	0.001192	0.001164	0.001293	0.000525	0.000514	0.000445	0.00085	0.000856	0.00102
4	3: Oats, sorghum and other cereal grains	0.024779	0.018949	1.020354	0.020555	0.021022	0.030835	0.02334	0.020534	0.025447	0.024588	0.027087	0.017839	0.017477	0.015152	0.010243	0.010146	0.01206
5	4: Wheat	0.058693	0.044937	0.04543	1.048687	0.049783	0.090321	0.055311	0.048706	0.060316	0.05831	0.064282	0.007452	0.007293	0.006312	0.011159	0.01128	0.01330
6	5: Barley	0.008322	0.0065	0.006918	0.006925	1.007057	0.07909	0.007834	0.006911	0.008545	0.008277	0.009113	0.024253	0.023775	0.020731	0.004106	0.003811	0.00453
7	6: Rice	0.000132	0.000115	0.000108	0.00011	0.000112	1.007598	0.000125	0.00011	0.000136	0.000132	0.000145	0.000155	0.000152	0.000131	6.03E-05	5.9E-05	7.04E-0
8	7: Oilseeds	0.010294	0.007841	0.008028	0.008527	0.008712	0.003471	1.009673	0.008519	0.010565	0.010201	0.011239	0.00165	0.001611	0.001384	0.002376	0.002392	0.00284
9	8: Legumes	0.01703	0.013049	0.012817	0.01412	0.014444	0.038264	0.016044	1.014111	0.017493	0.016892	0.018607	0.008797	0.008606	0.007453	0.001817	0.001717	0.00204
10	9: Beef cattle	0.007169	0.005528	0.006415	0.006002	0.006106	0.004113	0.006778	0.005992	1.007372	0.007172	0.007934	0.005496	0.005379	0.004647	0.005222	0.005225	0.00623
11	10: Untreated milk	0.002229	0.001722	0.002094	0.001868	0.0019	0.002175	0.002109	0.001865	0.002294	1.002232	0.002473	0.002314	0.002265	0.001958	0.002064	0.002067	0.0024
12	11: Dairy cattle	0.00042	0.000325	0.000395	0.000352	0.000358	0.00031	0.000398	0.000351	0.000432	0.000421	1.000466	0.000262	0.000256	0.000222	0.000387	0.000389	0.00046
13	12: Pigs	0.000441	0.000341	0.000423	0.00037	0.000376	0.000405	0.000417	0.00037	0.000454	0.000442	0.000491	1.000602	0.000589	0.000508	0.000447	0.000446	0.00053
14	13: Poultry	0.000551	0.000425	0.000532	0.000461	0.000469	0.000488	0.000521	0.000462	0.000567	0.000552	0.000613	0.00755	1.007408	0.006424	0.000678	0.000577	0.00068
15	14: Eggs	0.00016	0.000124	0.000163	0.000135	0.000137	9.14E-05	0.000152	0.000135	0.000165	0.000161	0.000179	0.000119	0.000116	1.000101	0.000199	0.0002	0.0002
16	15: Vegetables	0.002004	0.001553	0.002912	0.001685	0.001711	0.001515	0.001899	0.00168	0.002062	0.002013	0.002232	0.002592	0.00253	0.002171	1.006542	0.006582	0.0078
17	16: Fruit	0.001217	0.000942	0.001243	0.001022	0.001039	0.000692	0.001153	0.001019	0.001252	0.001222	0.001355	0.000804	0.000787	0.000682	0.00155	1.001561	0.00186
18	17: Plant nurseries	0.001111	0.000861	0.001913	0.000934	0.000949	0.000597	0.001053	0.000934	0.001144	0.001117	0.00124	0.000751	0.000735	0.000636	0.004803	0.004864	1.00582
19	18: Flowers	0.000324	0.000251	0.000547	0.000272	0.000277	0.00019	0.000307	0.000272	0.000333	0.000326	0.000362	0.000221	0.000216	0.000187	0.001364	0.001382	0.00165
20	19: Grapes for wine	0.001117	0.000864	0.001131	0.000938	0.000953	0.000597	0.001058	0.000939	0.00115	0.001121	0.001246	0.000671	0.000657	0.000569	0.001372	0.001384	0.00165
21	20: Horses	0.000481	0.000373	0.000804	0.000405	0.000411	0.000236	0.000456	0.000405	0.000495	0.000484	0.000537	0.022477	0.022087	0.019289	0.002327	0.002025	0.00242
22	21: Deer	2.47E-05	1.92E-05	4.12E-05	2.08E-05	2.11E-05	1.45E-05	2.34E-05	2.08E-05	2.54E-05	2.49E-05	2.76E-05	6.26E-05	6.14E-05	6.05E-05	0.000103	0.000105	0.00012
23	22: Animal breeding	0.001142	0.000888	0.001077	0.000964	0.000975	0.000552	0.001083	0.000957	0.001175	0.001148	0.001267	0.014694	0.014422	0.012555	0.001283	0.001083	0.00129
24	23: Sugar cane	0.00086	0.000669	0.001456	0.000723	0.000734	0.002362	0.000815	0.000722	0.000885	0.000864	0.000958	0.002073	0.002024	0.001736	0.003662	0.003689	0.00440
25	24: Unginned cotton	0.007433	0.005774	0.007415	0.006264	0.006353	0.002356	0.00705	0.006343	0.007659	0.007498	0.008404	0.00425	0.004144	0.003534	0.008638	0.008708	0.01042
26	25: Hops	0.000108	8.44E-05	0.000101	9.13E-05	9.24E-05	0.00017	0.000103	9.11E-05	0.000111	0.000109	0.000121	0.000323	0.000315	0.000271	0.000103	1E-04	0.00011
27	26: Grass seed	0.002145	0.00165	0.002918	0.001793	0.001824	0.000705	0.002024	0.001794	0.002206	0.002146	0.002385	0.001429	0.001399	0.001213	0.005966	0.006035	0.00723
28	27: Hay	0.048079	0.037505	0.041673	0.040669	0.04116	0.015061	0.04567	0.040349	0.049501	0.048495	0.053599	0.031083	0.030449	0.026406	0.028281	0.028249	0.03383
29	28: Natural rubber	0.002159	0.001662	0.00206	0.001806	0.001836	0.00123	0.002038	0.001801	0.00222	0.002161	0.002395	0.006487	0.006358	0.005508	0.002255	0.002195	0.00262
30	29: Skins and other agricultural services	0.077487	0.06	0.08198	0.065147	0.066181	0.023419	0.073431	0.06489	0.079759	0.077868	0.086255	0.050887	0.049863	0.04328	0.110688	0.111598	0.13371

... | IO_345 demand | IO_345 table | Validation | X | inverse | A coefficients | L matrix | q matrix | m matrix ...

S5. Emissions intensities and footprint

Actual MMULT formulas for q, m and induced output, selected contributions and total footprint. Worksheet snapshots from Microsoft Excel.

B2 =MMULT(Satellite!A1:MG1,'X ^ inverse'!A1:MG345)

q = MMULT(Q, Xhat inverse); kt CO2-e per AUD million	1: Sheep a	2: Shorn w	3: Oats, so	4: Wheat	5: Barley	6: Rice	7: Oilseeds	8: Legume	9: Beef cat	10: Untrea	11: Dairy c	12: Pigs	13: Poultry	14: Eggs	15: Vegeta	16: Fruit	17
	2.3546	2.799302	0.237653	0.238787	0.171309	0.4575	0.170215	0.95512	5.283126	2.069473	0.394962	1.610806	0.495584	0.315734	0.225814	0.229221	0

Sheet tabs: ... | X ^ | X ^ inverse | A coefficients | L matrix | q matrix | m matrix | Footprint sectors | Footprint summary | Fc ...

B2 =MMULT('q matrix'!B2:MH2,'L matrix'!B2:MH346)

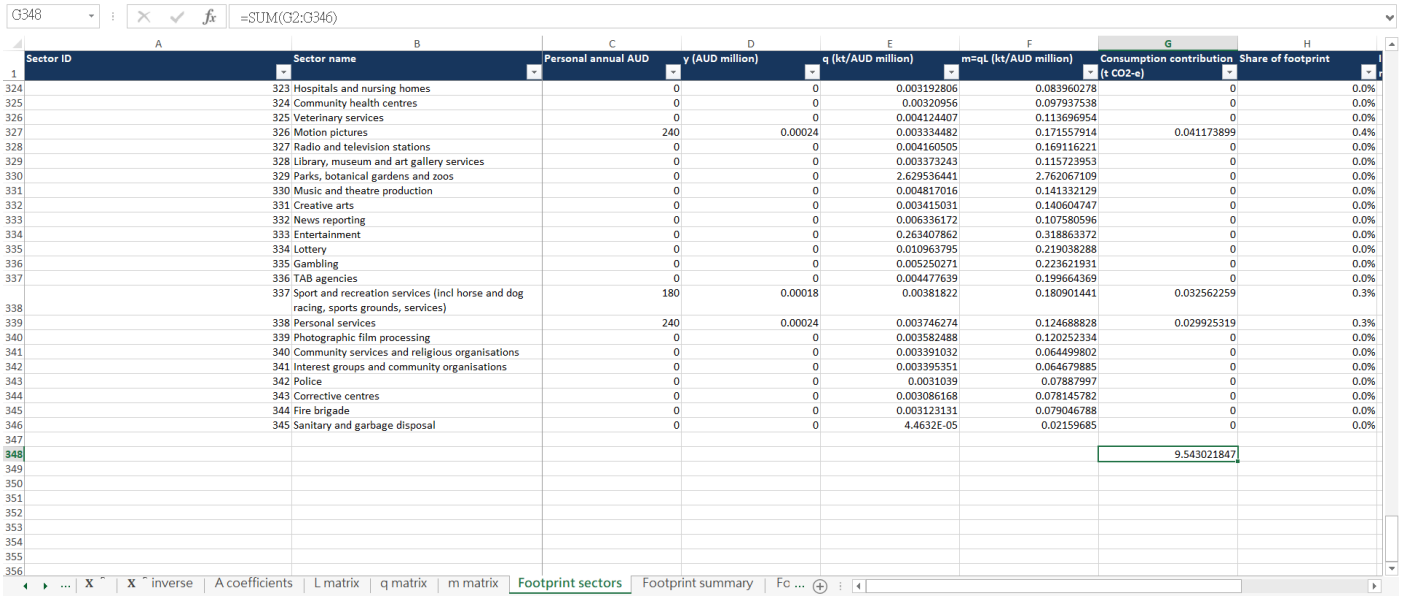
m = q * L (kt/AUD million)	1: Sheep a	2: Shorn w	3: Oats, so	4: Wheat	5: Barley	6: Rice	7: Oilseeds	8: Legume	9: Beef cat	10: Untrea	11: Dairy c	12: Pigs	13: Poultry	14: Eggs	15: Vegeta	16: Fruit	17: Pla
	2.663759	3.038517	0.525606	0.498051	0.434816	0.774417	0.462837	1.216099	5.601436	2.379646	0.740981	1.864843	0.743759	0.529671	0.499417	0.503669	0.553

Sheet tabs: ... | X ^ | X ^ inverse | A coefficients | L matrix | q matrix | m matrix | Footprint sectors | Footprint summary | Fc ...

G16 =F16*D16*1000

Sector ID	Sector name	Personal annual AUD	y (AUD million)	q (kt/AUD million)	m=qL (kt/AUD million)	Consumption contribution (t CO2-e)	Share of footprint
1	Sheep and lambs	0	0	2.354599675	2.663758633	0	0.0%
2	Shorn wool	0	0	2.799302305	3.038517132	0	0.0%
3	Oats, sorghum and other cereal grains	0	0	0.237652511	0.525605901	0	0.0%
4	Wheat	0	0	0.23878743	0.498050711	0	0.0%
5	Barley	0	0	0.171308859	0.434815745	0	0.0%
6	Rice	0	0	0.457499681	0.774417269	0	0.0%
7	Oilseeds	0	0	0.170214688	0.462836504	0	0.0%
8	Legumes	0	0	0.955120403	1.216099165	0	0.0%
9	Beef cattle	0	0	5.283125771	5.601436209	0	0.0%
10	Untreated milk	0	0	2.069473482	2.379645651	0	0.0%
11	Dairy cattle	0	0	0.394961672	0.740980845	0	0.0%
12	Pigs	0	0	1.610805662	1.864843091	0	0.0%
13	Poultry	0	0	0.495583696	0.743758842	0	0.0%
14	Eggs	0	0	0.315733616	0.529671286	0	0.0%
15	Vegetables	720	0.00072	0.225813873	0.499416931	0.35958019	3.8%
16	Fruit	480	0.00048	0.229220827	0.503668563	0.24176091	2.5%
17	Plant nurseries	0	0	0.226348648	0.553743309	0	0.0%
18	Flowers	0	0	0.299180042	0.667600949	0	0.0%
19	Grapes for wine	0	0	0.230505023	0.630730109	0	0.0%
20	Horses	0	0	0.404678105	0.658301575	0	0.0%
21	Deer	0	0	0.210835394	0.355642696	0	0.0%
22	Animal breeding	0	0	0.430194171	0.679805922	0	0.0%
23	Sugar cane	0	0	0.666437607	0.961926773	0	0.0%
24	Unginned cotton	0	0	0.207382791	0.48939276	0	0.0%
25	Hops	0	0	0.067463782	0.230071822	0	0.0%
26	Grass seed	0	0	0.786372871	0.987779514	0	0.0%
27	Hay	0	0	0.161423554	0.40334478	0	0.0%
28	Natural rubber	0	0	0.164244865	0.346422076	0	0.0%
29	Skins and other agricultural services	0	0	0.2405628	0.447708595	0	0.0%
30	Ginned cotton	0	0	0.169503132	0.366633004	0	0.0%
31	Cotton seed	0	0	0.169503132	0.366633004	0	0.0%
32	Sheep shearing	0	0	0.066625788	0.28914761	0	0.0%
33	Aerial agriculture	0	0	0.011871927	0.159695223	0	0.0%
34	Forest products	0	0	0.078312236	0.247476905	0	0.0%

Sheet tabs: ... | X ^ | X ^ inverse | A coefficients | L matrix | q matrix | m matrix | Footprint sectors | Footprint summary | Fc ...



G348 =SUM(G2:G346)

	Sector ID	Sector name	Personal annual AUD	y (AUD million)	q (kt/AUD million)	m=qL (kt/AUD million)	Consumption contribution (t CO2-e)	Share of footprint
324	323	Hospitals and nursing homes	0	0	0.003192806	0.083960278	0	0.0%
325	324	Community health centres	0	0	0.00320956	0.097937538	0	0.0%
326	325	Veterinary services	0	0	0.004124407	0.113696954	0	0.0%
327	326	Motion pictures	240	0.00024	0.003334482	0.171557914	0.041173899	0.4%
328	327	Radio and television stations	0	0	0.004160505	0.169116221	0	0.0%
329	328	Library, museum and art gallery services	0	0	0.003373243	0.115723953	0	0.0%
330	329	Parks, botanical gardens and zoos	0	0	2.629536441	2.762067109	0	0.0%
331	330	Music and theatre production	0	0	0.004817016	0.141332129	0	0.0%
332	331	Creative arts	0	0	0.003415031	0.140604747	0	0.0%
333	332	News reporting	0	0	0.006336172	0.107580596	0	0.0%
334	333	Entertainment	0	0	0.263407862	0.318863372	0	0.0%
335	334	Lottery	0	0	0.010963795	0.219038288	0	0.0%
336	335	Gambling	0	0	0.005250271	0.223621931	0	0.0%
337	336	TAB agencies	0	0	0.004477639	0.199664369	0	0.0%
338	337	Sport and recreation services (incl horse and dog racing, sports grounds, services)	180	0.00018	0.00381822	0.180901441	0.032562259	0.3%
339	338	Personal services	240	0.00024	0.003746274	0.124688828	0.029925319	0.3%
340	339	Photographic film processing	0	0	0.003582488	0.120252334	0	0.0%
341	340	Community services and religious organisations	0	0	0.003391032	0.064499802	0	0.0%
342	341	Interest groups and community organisations	0	0	0.003395351	0.064679885	0	0.0%
343	342	Police	0	0	0.0031039	0.07887997	0	0.0%
344	343	Corrective centres	0	0	0.003086168	0.078145782	0	0.0%
345	344	Fire brigade	0	0	0.003123131	0.079046788	0	0.0%
346	345	Sanitary and garbage disposal	0	0	4.4632E-05	0.02159685	0	0.0%
347								
348							9.543021847	
349								
350								
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355								
356								

... | X ^ | X ^ inverse | A coefficients | L matrix | q matrix | m matrix | Footprint sectors | Footprint summary | Fo ...

Worksheet snapshots rendered by LibreOffice Calc from Labels.xlsx.

S5. q, m and footprint calculation

Sources: q matrix!B2:MH2, m matrix!B2:MH2, Footprint sectors and Footprint summary.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Step	Source	Exact formula / units
q	'q matrix'!B2:MH2	Formula: =MMULT(Satellite!A1:MG1,'$\hat{\mathbf{X}}$ inverse'!A1:MG345)
m	'm matrix'!B2:MH2	Formula: =MMULT('q matrix'!B2:MH2,'L matrix'!B2:MH346)
Contribution	'Footprint sectors'!G2:G346	Formula: G_i = m_i * y_i * 1000 (tonnes CO2-e)
Induced output	'Footprint sectors'!I2	Formula: =MMULT('L matrix'!B2:MH2,$D$2:$D$346)
Total	'Footprint summary'!B3	Formula: =SUM('Footprint sectors'!G2:G346)

Sector ID	Purchase sector	q (kt/AUD m)	m (kt/AUD m)	Annual AUD	t CO2-e/year
75	Fresh meat	0.091	2.376	720.000	1.711
79	Treated milk	2.522	2.727	480.000	1.309
241	Electricity supply	1.380	2.456	480.000	1.179
77	Poultry, slaughtered	0.218	2.075	480.000	0.996
259	Hotels, clubs, restaurants and cafes	0.003	0.209	2,400.000	0.502
265	Railway passenger transport services	0.317	0.455	1,080.000	0.491
292	Ownership of dwellings	0.000	0.018	15,600.000	0.278
	TOTAL t CO2-e/person/year	9.543			

Multiplication uses compatible dimensions: q (1×345), L (345×345), y (345×1). qLy yields kt; multiply by 1,000 for tonnes.

S6. Consumption category results

Top ten consumption categories. Archived worksheet view captured on 7 November 2026 using LibreOffice Calc; the national comparison is presented in Figure 2.

S7. Provisional footprint and sector hotspots

Source: Footprint summary!A2:D22. Consumption categories driving emissions; not emitting-sector locations.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Rank	Consumption category	t CO2-e/year	Share
1.000	Fresh meat	1.711	17.9%
2.000	Treated milk	1.309	13.7%
3.000	Electricity supply	1.179	12.4%
4.000	Poultry, slaughtered	0.996	10.4%
5.000	Hotels, clubs, restaurants and cafes	0.502	5.3%
6.000	Railway passenger transport services	0.491	5.1%
7.000	Vegetables	0.360	3.8%
8.000	Ownership of dwellings	0.278	2.9%
9.000	Clothing	0.265	2.8%
10.000	Cheese	0.256	2.7%
	Total	9.543	

PRELIMINARY: nominal expenditure × 2018 intensities. Price basis, purchaser/basic-price valuation, insurance and direct household combustion remain unresolved. Australian/international comparisons are not yet available.

S7. Food substitution scenario

Source: Labels.xlsx, Reduction scenario worksheet. All other expenditure and model intensities remain unchanged. The six affected sectors and reconciled totals are shown below.

Sector	Baseline AUD	Change AUD	Revised AUD
Legumes	0	+192	192
Vegetables	720	+192	912
Fresh meat	720	-144	576
Poultry, slaughtered	480	-96	384
Treated milk	480	-96	384
Cheese	240	-48	192
Annual budget	29,760	0	29,760

Baseline: 9.543 t CO_2e/year. Scenario: 9.018 t CO_2e/year. Difference: 0.525 t CO_2e/year (5.50%).